

Problem

Design a problem to help other students better understand how to correct power factor to values other than unity.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand how to correct power factor to values other than unity.

Although there are many ways to work this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

A 120-V rms, 60-Hz electric hair dryer consumes 600 W at a lagging pf of 0.92. Calculate the rms-valued current drawn by the dryer.

How would you power factor correct this to a value of 0.95?

Solution

$$P = 600 \text{ W}, \text{ pf} = 0.92 \longrightarrow \theta = 23.074^\circ$$

$$P = S \cos \theta \longrightarrow S = \frac{P}{0.92} = 652.17 \text{ VA}$$

$$S = P + jQ = 600 + j652.17 \sin 23.09^\circ = 600 + j255.6$$

$$\text{But } S = V_{\text{rms}} I_{\text{rms}}^*$$

$$I_{\text{rms}}^* = \frac{S}{V_{\text{rms}}} = \frac{600 + j255.6}{120}$$

$$I_{\text{rms}} = 5 - j2.13 = \mathbf{5.435 \angle -23.07^\circ \text{ A}}$$

To correct this to a pf = 0.95, I would add a capacitor in parallel with the hair dryer (remember, series compensation will increase the power delivered to the load and probably burn out the hair dryer.

$$\text{pf} = 0.95 = 600/S \text{ or } S = 631.6 \text{ VA and } \theta = 18.19^\circ \text{ and VARs} = 197.17$$

Thus,

$$\text{VAR}_{\text{scap}} = 255.6 - 197.17 = 58.43 = 120 \times I_C \text{ or } I_C = 58.43/120 = 0.4869 \text{ A}$$

Next,

$$X_C = 120/0.4869 = 246.46 = 1/(377 \times C) \text{ or } C = \mathbf{10.762 \mu\text{F}}$$